

SEAT NO. 32

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
THIRD YEAR (BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2023
Batch 2020

Time : 3 Hours

Dated : 31-JUL-23
 Max Marks : 60

Computer Communication Networks - CT-376

- Read the questions carefully.
- Attempt all questions.
- All questions carry equal marks.

Q#1 Explain the process of data transmission through the layers of the OSI Model, highlighting the responsibilities and functions of each layer. Identify and discuss at least two challenges that may arise during data transmission and how they can be mitigated. Provide examples or scenarios to support your explanation. [CLO-1]

Q#2 Explain and highlight the advantages and disadvantages of the following network topologies:

- A. Bus Topology
- B. Star Topology
- C. Ring Topology
- D. Mesh Topology
- E. Tree Topology

Note: Explain each topology with the help of diagrams.

[CLO-1]

Q#3 Explain the followings with the help of examples:

- A. Circuit Switching and Packet Switching Techniques.
- B. Carrier Sense Multiple Access with Collision Detection (CSMA / CD)
- C. Static and Dynamic Routing Protocols

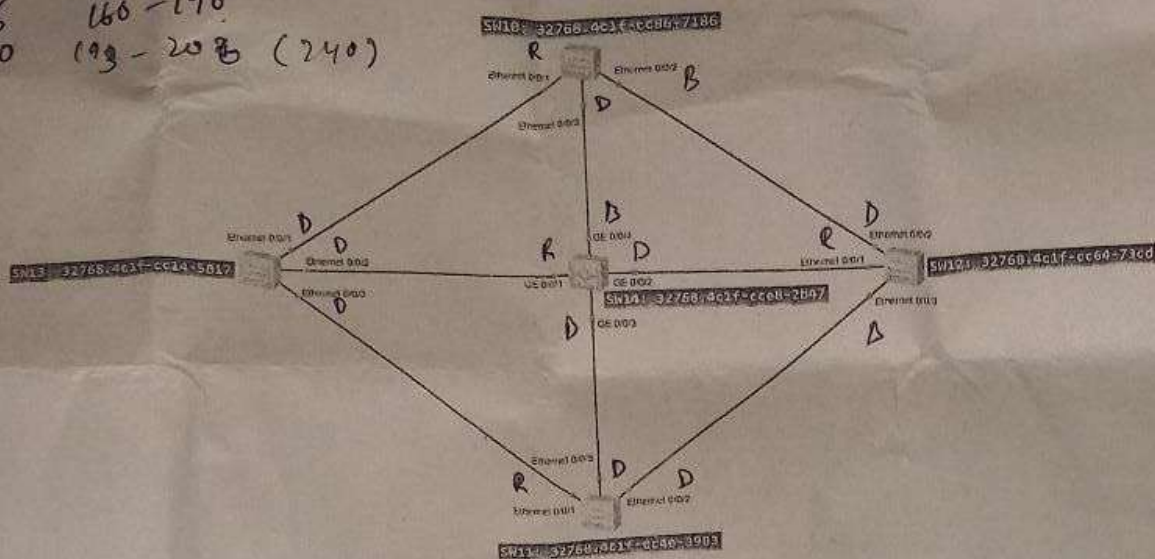
[CLO-1]

Q#4 Determine the Root Ports (RP), Designated Ports (DP) and Alternate Ports (AP) in the following network topology when STP is fully converged:

Note: Provide explanation of the process involved in identifying each of the aforementioned ports.

[CLO-2]

60 0-62 (192)
 48 65-126 (192)
 24 129-158 (224)
 15 160-190 (224)
 10 193-208 (240)



Switch Information

SW10: 32768.4c1f-cc86-7186
 SW11: 32768.4c1f-cc4e-3983
 SW12: 32768.4c1f-cc64-73cd
 SW13: 32768.4c1f-cc14-5817
 SW14: 32768.4c1f-cce8-2b47

- #5 You are tasked with designing a highly scalable network infrastructure for a rapidly growing organization with the IP address range 192.168.0.0/24. The organization has multiple departments with varying requirements for subnet sizes and host addresses. Additionally, the network needs to accommodate future expansion and ensure efficient utilization of IP addresses.

Using Variable Length Subnet Masking (VLSM) technique, design a sub-netting plan that meets the following requirements:

- Subnet 1: Sales Department - requires a minimum of 50 host addresses.
- Subnet 2: Marketing Department - requires a minimum of 40 host addresses.
- Subnet 3: Research Department - requires a minimum of 20 host addresses.
- Subnet 4: IT Department - requires a minimum of 12 host addresses.
- Subnet 5: Administration Department - requires a minimum of 08 host addresses.

Furthermore, considering future expansion, each subnet should be able to accommodate an additional 20% growth in the number of hosts.

- a) Calculate the subnet mask for each subnet, and provide the range of usable IP addresses for each subnet.
- b) Determine the network address and broadcast address for each subnet.

Note: Take the future expansion into account and ensure efficient utilization of IP addresses in your design.

[CLO-2]

SEAT NO. CT-21030

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
THIRD YEAR(Computer Science)
SPRING SEMESTER EXAMINATIONS 2024

Batch 2021

Dated : 19-JUL-24

Max Marks : 60

Time : 3 Hours

Computer Communication Networks - CT-376

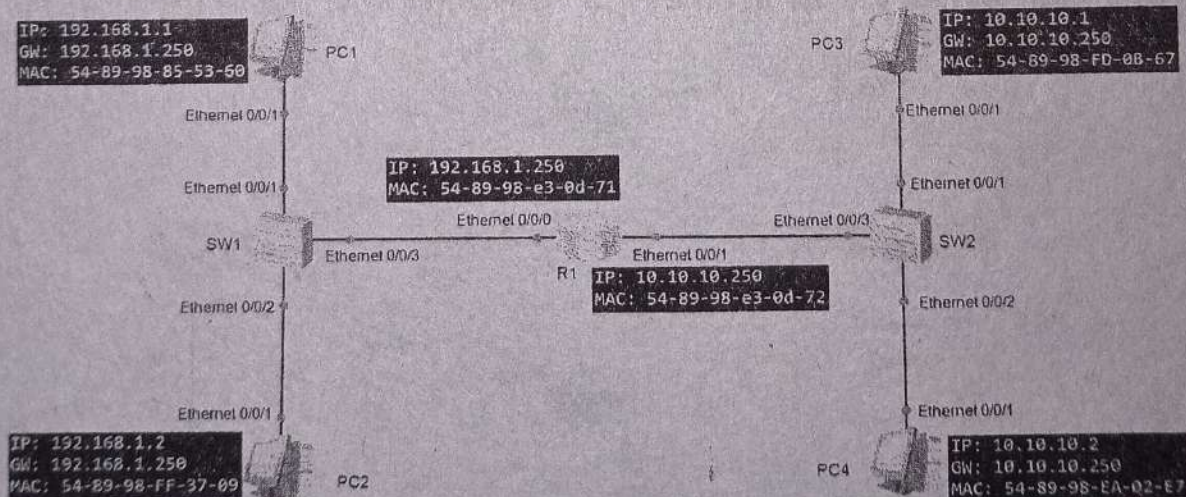
- Read the questions carefully.
- Attempt all questions.
- All questions carry equal marks.

Q#1 **Explain** the Network Address Translation (NAT) and its types with the help of examples. [CLO-1]

Q#2 **Explain** the Domain Name System (DNS) and its functionality with an example. Why is DNS necessary? Discuss the importance of Root and Top-Level Domains (TLDs). Also, explain the difference of private and public DNS. [CLO-1]

Q#3 In the given network topology, **explain** the packet flow process for the following two scenarios:

- PC1 attempts to communicate with PC2.
- PC2 attempts to communicate with PC4.



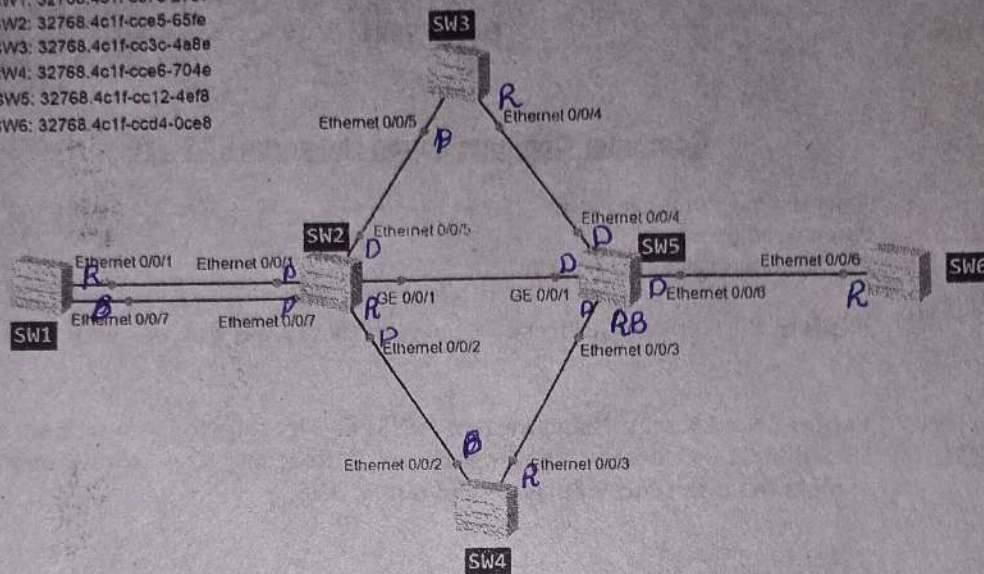
For each scenario, provide the ARP table and MAC address table (where applicable) for each device involved in the communication.

[CLO-1]

- Q#4 Determine the Root Ports (RP), Designated Ports (DP) and Alternate Ports (AP) in the following network topology when STP is fully converged:

Switch numbers with their BIDs

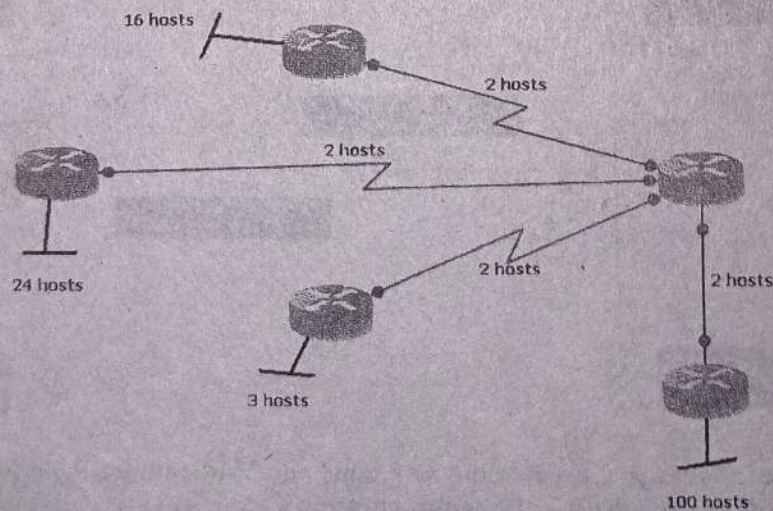
SW1: 32768.4c1f-cc7c-2757
 SW2: 32768.4c1f-cc65-65fe
 SW3: 32768.4c1f-cc3c-4a8e
 SW4: 32768.4c1f-cc66-704e
 SW5: 32768.4c1f-cc12-4e78
 SW6: 32768.4c1f-ccd4-0ce8



Note: Provide explanation of the process involved in identifying each of the aforementioned ports.

[CLO-2]

- Q#5 Using Variable Length Subnet Masking (VLSM) on available network 84.83.85.0/24, design a sub-netting plan that meets the following requirements:



Calculate the subnet mask for each subnet, and provide the range of usable IP addresses for each subnet. Also, calculate the network address and broadcast address for each subnet.

Note: Show all steps of calculations.

[CLO-2]

ENGINEERING & TECHNOLOGY
THIRD YEAR (Computer Science)
SPRING SEMESTER EXAMINATIONS 2025

Time : 3 Hours

Batch 2022

Dated : 27-MAY-25

Max Marks : 60

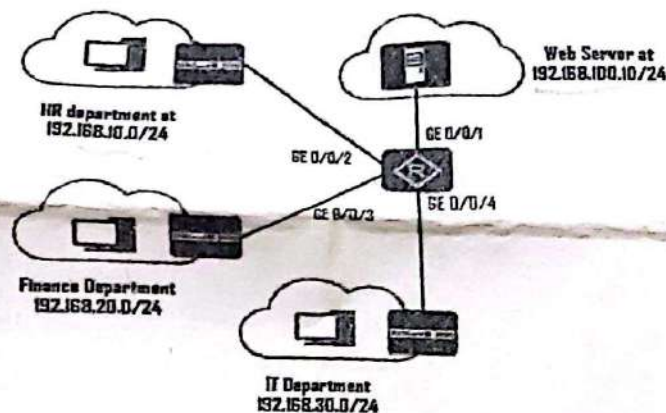
Dr. Shariq

Computer Communication Networks - CT-376

Instructions:

- Attempt all questions. Each question carries equal marks.
- You may attempt the question in any order; however, the parts of the question must be answered in sequence.

Q.1(a) You are a network administrator for a company with the following network setup: (06)
CLO-2



Construct an Extended ACL on the router to meet the following security policies:

- Permit HTTP access to the Web Server only from the HR Department.
- Block all access to the Web Server from the Finance Department.
- Allow HTTP access to the Web Server from the IT Department.
- Deny all other unauthorized traffic to the Web Server.

Determine the correct interface and direction (inbound/outbound) to apply the ACL for optimal traffic filtering. Apply the ACL to the router's configuration.

- (b) Compare the manual and LACP modes of Eth-Trunk in terms of configuration complexity, fault detection, and scalability. Identify why LACP mode is preferred in enterprise networks despite its higher complexity. (06)
CLO-2

- Q.2(a) You are a network engineer at XYZ Corporation tasked to **build** an efficient IP addressing scheme for a new branch office. The office has the following departments with specific host requirements: (06) CLO-2

Administration:	60 hosts
Sales:	30 hosts
IT Support:	12 hosts
Guest Wi-Fi:	5 hosts

The company has been allocated the network 192.168.100.0/24. Provide the subnet details for each department which includes the network address, broadcast address, and range of available addresses for each subnet.

- (b) **Apply** the concepts of trunking and VLAN tagging to explain how data from different VLANs can traverse a single physical link between a switch and a router. (06) CLO-2

- Q.3(a) **Discuss** the steps of the TCP three-way handshake for connection establishment and describe the process of data encapsulation and decapsulation in the TCP/IP model. (06) CLO-1

- (b) **Distinguish** between star and ring topologies in terms of structure and fault tolerance and Compare the characteristics of LAN, MAN, and WAN in terms of coverage and typical use cases. (06) CLO-1

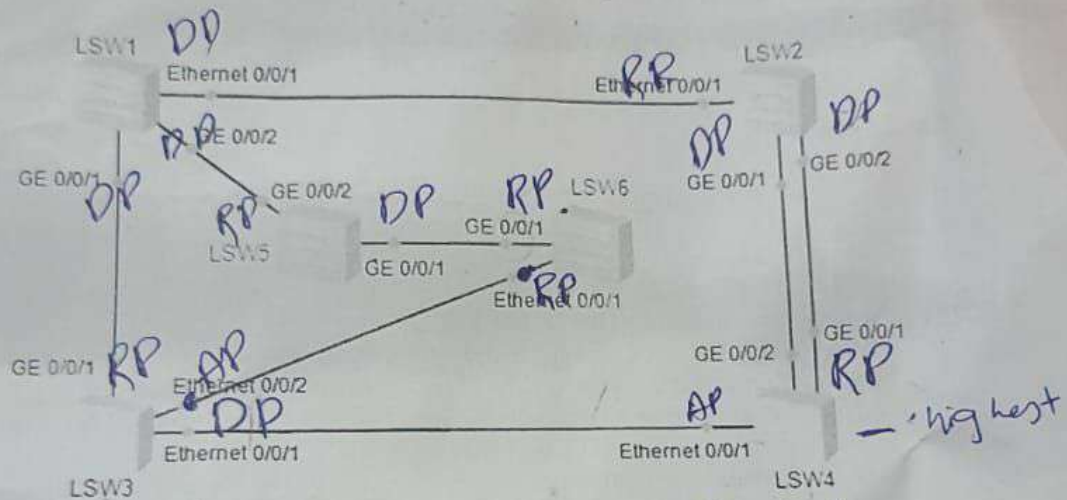
- Q.4(a) Describe the key differences between static and dynamic routing and **outline** the steps a router takes to forward a packet using its routing table with the help of real life example. (06) CLO-1

- (b) **Compare** distance-vector and link-state routing protocols in terms of route advertisement and network visibility and describe the purpose of the Link State Database (LSDB) in the OSPF protocol. (06) CLO-1

- Q.5(a) Describe the differences between Access, Trunk, and Hybrid interfaces in VLAN implementation and **discuss** interface-based VLAN assignment and MAC-based VLAN assignment. (06) CLO-1

- (b) **Explain** the primary purpose of the Spanning Tree Protocol (STP) in Ethernet networks with the help of an example and describe the election process of the root bridge in STP. (06) CLO-1

Q#4 Determine the Root Ports (RP), Designated Ports (DP) and Alternate Ports (AP) in the following network topology when STP is fully converged:



LSW1 :	32768.4c1f-cc03-6d96	- Root & S
LSW2 :	32768.4c1f-cc99-4c72	
LSW3 :	32768.4c1f-cc37-5d3e	
LSW4 :	32768.4c1f-ccd3-7ba4	- highest
LSW5 :	8192.4c1f-cc08-572b	
LSW6 :	32768.4c1f-cc20-7c7b	

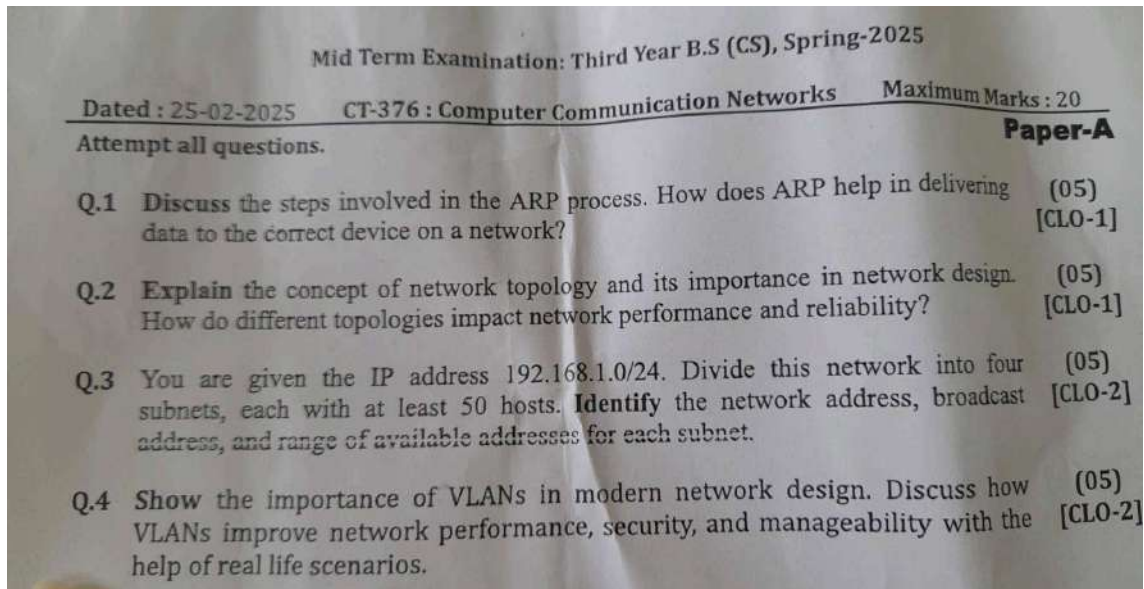
Provide explanation of the STP process involved in identifying each of the aforementioned ports.

[CLO-2] - (10 Marks)

25/02/2025 Tuesday

CCN

Prof. Shariq Khan



Paper B

Q1) TCP/IP & OSI model compare and contrast. And why TCP is widely used in practice ?

Q2) 3 Network Topologies Specific Ring, Star, Tree.
Compare, contrast, advantages, disadvantages.

Q3) IP 172.16.0.0/16 given. Do subnetting and divide it for Department A 200 Hosts, Department B 100 Hosts and Department C 50 Hosts. Write Network IDs, range of available addresses and broadcast IPs for each subnet.

Q4) Vlan assignment methods. Compare, contrast, advantages, disadvantages. According to your understanding, which method is best for the scenario where device is frequently moved through different network ports?

Instructor: Sir Karim Kazi

Paper A

Q1.

Explain the OSI model in detail, including the function of each layer. Also explain why the ARP is used in networking. Support your answer with a suitable **network topology diagram**.

(Paper B had TCP/IP model in detail with other topic here)

Q2.

Define VLAN with its different assignment types. Also, outline the steps involved in VLAN assignment in a network, supported by appropriate network topology diagrams.

Q3.

Given the following IP configuration:

- IP Address: **172.16.14.37**
- Subnet Mask: **255.255.248.0**

Answer the following:

1. Determine the network address, broadcast address, and range of usable IP addresses in this subnet
2. List all possible IP addresses that belong to this network.
3. Can 172.16.9.1 be assigned as the **default gateway** for this network? Justify your answer.